



CALAYAN EDUCATIONAL FOUNDATION, INC.

COLLEGE OF BUSINESS AND MANAGEMENT



INCUBATION **PROJECT**

CEFI Facebook Messenger-Based Facility Reservation System with AI Automation using n8n

Calayan Educational Foundation, Inc.
Business Address

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Project Consultant

A.Y. 2025-2026 | 2nd Semester



I. Client & Business Profile

a. Client Company Background

- Calayan Educational Foundation, Inc. (CEFI) is an educational institution that provides academic services to students and supports various school-related activities. The institution manages multiple facilities such as function halls, classrooms, and event spaces that are used for academic and non-academic purposes.

b. Nature of Business

- Educational Institution offering academic programs and facility services for events, meetings, and school-related functions.

c. Organizational Structure

- School Administrator
- Facility Manager / Admin Staff
- Reservation Personnel
- IT Support

d. Current Business Processes

- The current facility reservation process at CEFI is entirely manual and involves the following steps:
 - Clients (students, faculty, or external users) inquire about facility availability through:
 - Face-to-face communication
 - The staff manually checks availability using:
 - Logbooks
 - Paper schedules
 - Verbal coordination with other staff
 - Reservation details are written manually in logbooks or forms.
 - Staff confirms reservations manually by replying to the client.
 - Payment (if applicable) is processed separately and recorded manually.
 - There is **no centralized system**, leading to:
 - Double bookings
 - Lost or incomplete records
 - Delayed responses to inquiries
 - Difficulty tracking reservation history

e. Business Documents (DTI, COR, Permit)

-

f. Problem Statement

- The current reservation system is inefficient due to its manual nature.
- There is no automated way to handle inquiries and reservations.
- Staff experience delays in responding to clients.
- High risk of human error such as double booking and incorrect scheduling.
- Lack of centralized database for monitoring and reporting.
- No real-time availability checking for clients.



- g. Scope and Limitations
 - Scope
 - The system will automate facility reservation using Facebook Messenger.
 - It will integrate n8n workflows and AI agents for processing requests.
 - The system will manage:
 - Reservation requests
 - Availability checking
 - Automated responses
 - Data storage
 - Limitation
 - Requires internet connection for Messenger and automation tools.
 - Limited to users who can access Facebook Messenger.
 - Advanced payment integration may not be included.

II. Proposed IT Business Solution

- a. Overview of Proposed Solution
 - The proposed system is an **automated facility reservation system** that uses Facebook Messenger as the main communication platform. It integrates **n8n workflows and AI agents** to handle inquiries, process reservations, and manage scheduling automatically.

Instead of manual handling, the system will:

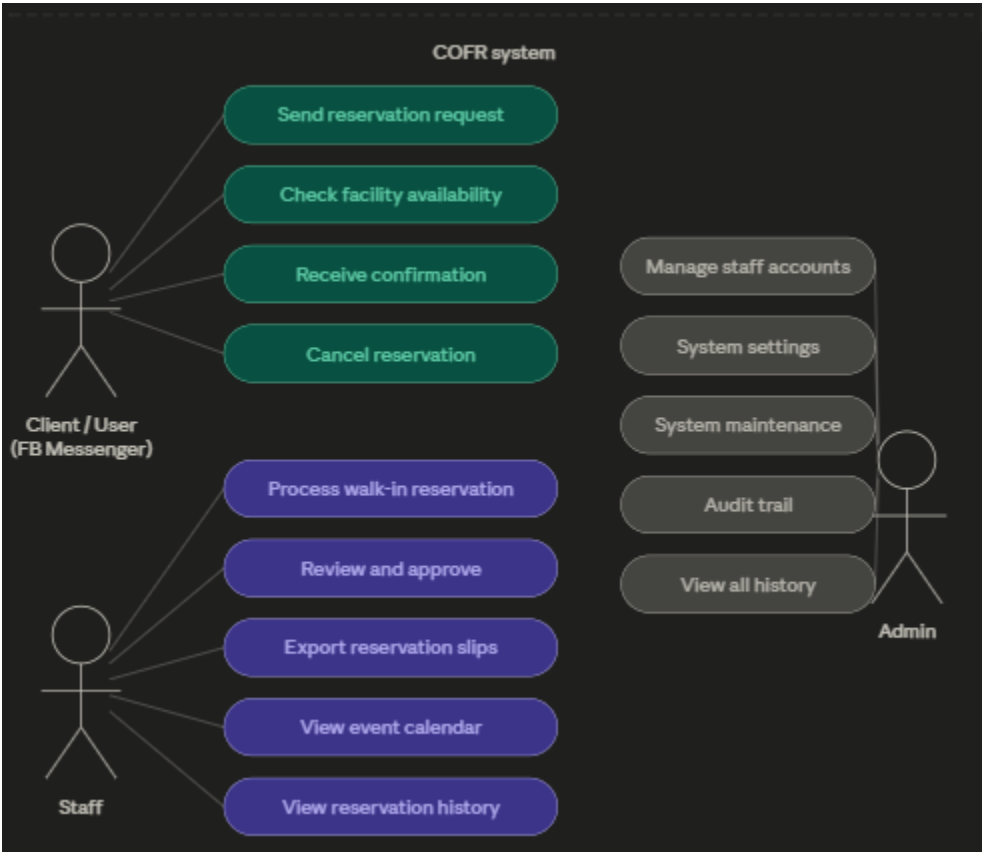
- Receive reservation requests via Messenger
- Process requests using AI agents
- Automatically check availability
- Store reservation data in a database
- Send confirmations instantly

Layer	Technology	Version	Purpose
Frontend	HTML5 / CSS3 / JavaScript	ES2022	Web interface for staff and admin dashboards
Backend	PHP	8.2	Server-side logic, API routing, session management
Database	MySQL	8.0	Stores reservations, facilities, users, audit logs
DB Management	phpMyAdmin	5.2	Database administration and query management
Automation	n8n	1.x (self-hosted)	Workflow automation engine connecting Messenger, AI agent, and COFR API

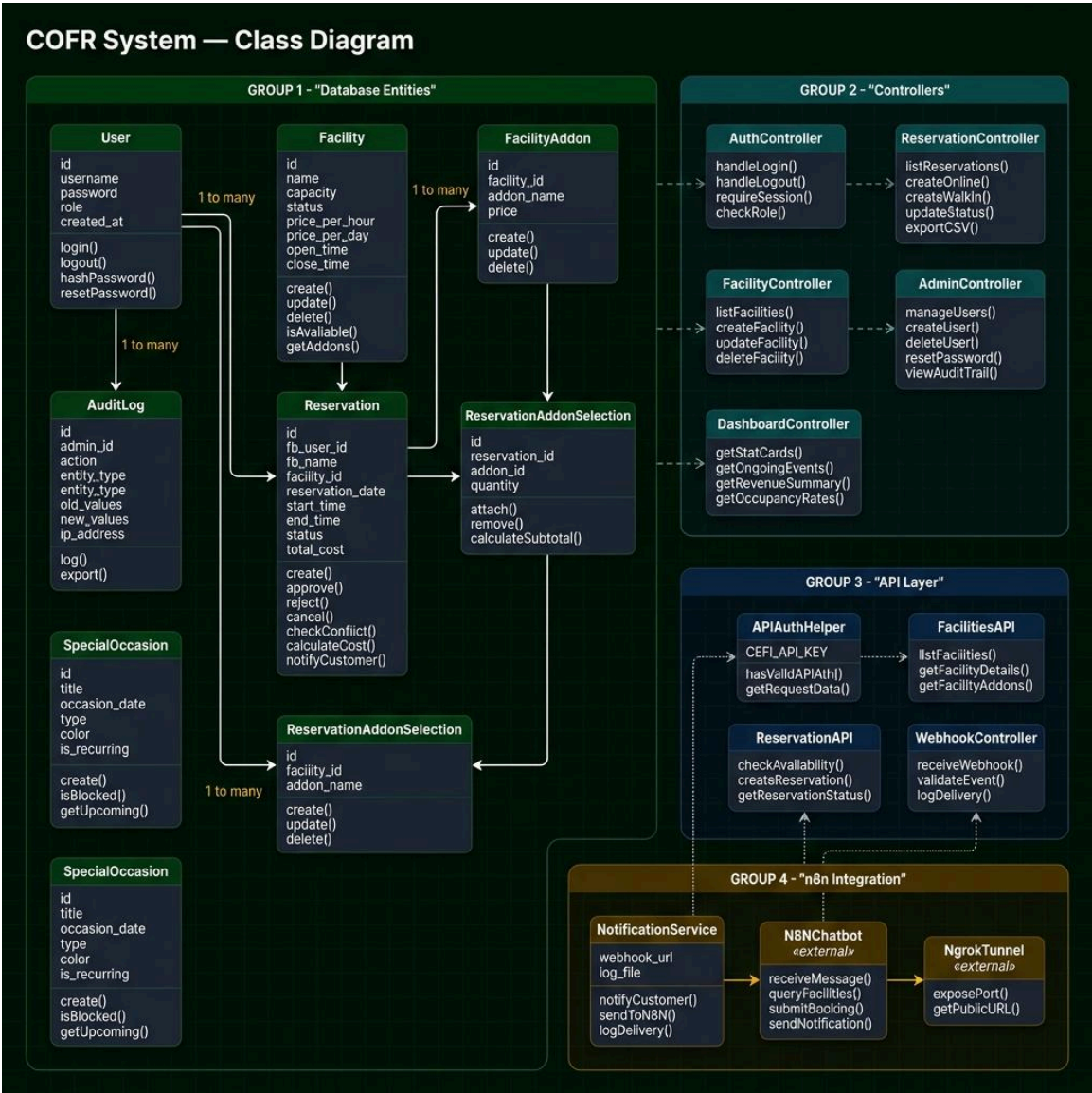


AI Agent	n8n AI Agent node		Processes natural language reservation requests from clients via Messenger
Messaging API	Facebook Messenger API (Meta Graph API)	v19.0	Receives and sends messages to/from clients on Facebook Messenger
Webhook	ngrok	3.x	Exposes local n8n instance to Meta webhook endpoint during development
Infrastructure	Apache (XAMPP)	2.4	Local web server hosting the COFR PHP application
Runtime	Node.js	20 LTS	Required runtime for n8n self-hosted instance

- b. Solution Objectives
 - To automate the facility reservation process
 - To reduce manual workload of staff
 - To provide real-time responses to clients
 - To eliminate double booking and errors
 - To centralize reservation data
 - To improve overall efficiency and customer satisfaction
- c. Proposed System Features
 - Facebook Messenger chatbot interface
 - AI-powered conversation handling
 - Automated reservation processing via n8n
 - Real-time availability checking
 - Automated confirmation and notifications
 - Reservation database management
 - Admin dashboard
 - Reporting and monitoring tools
- d. Use Case Diagram



e. Database Design (if applicable)



- III. Prototype / System Development Evidence
- a. Overview of the System: Short description of the system or solution; and Objective of the system.

The CEFI Online Facility Reservation (COFR) system is a web-based repository for managing institutional spaces (gyms, labs, etc.). It bridges the gap between online users (social media integration) and walk-in clients, providing administrators with real-time tools to track availability, revenue, and audit trails.

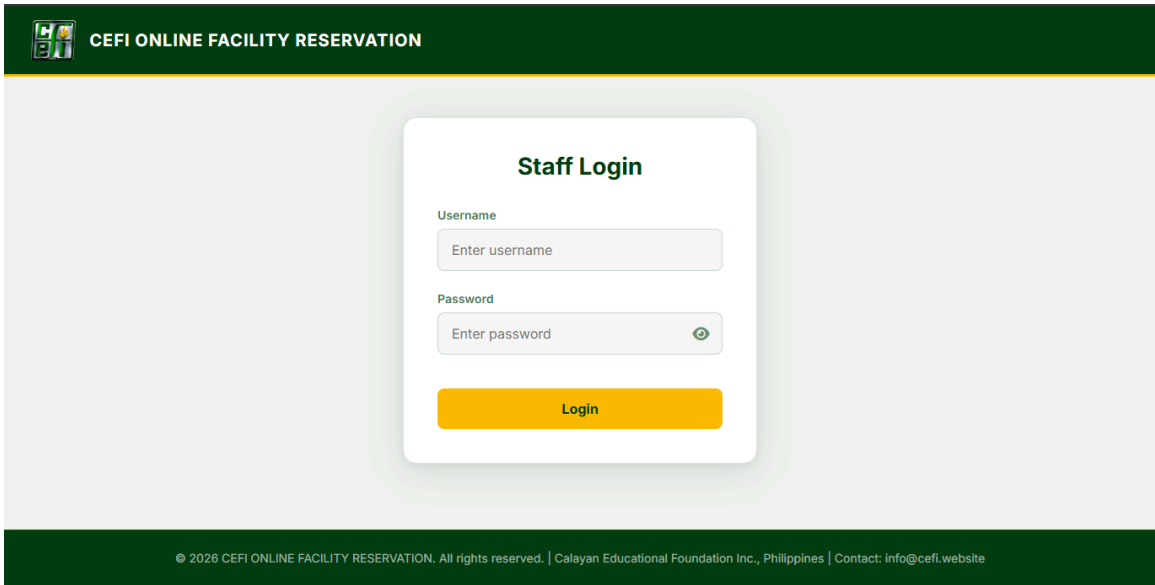
Key Objectives

- Centralized Management: Unified control for all facilities, pricing, and operating hours.
- Streamlined Workflow: Automated reservation status handling and requirements gathering (equipment needs).
- Conflict Resolution: Real-time calendar synchronization to prevent double-bookings.
- Financial Tracking: Dynamic reporting for daily, monthly, and total rental revenue.
- Security & Accountability: Granular audit logging for every administrative action.

- b. Screenshots of the System/Project
- Use captions below each screenshot to explain what it is and which module it belongs to.
 - If the system is interactive, consider including step-by-step screenshots of key processes (e.g., submitting a transaction, generating a report).

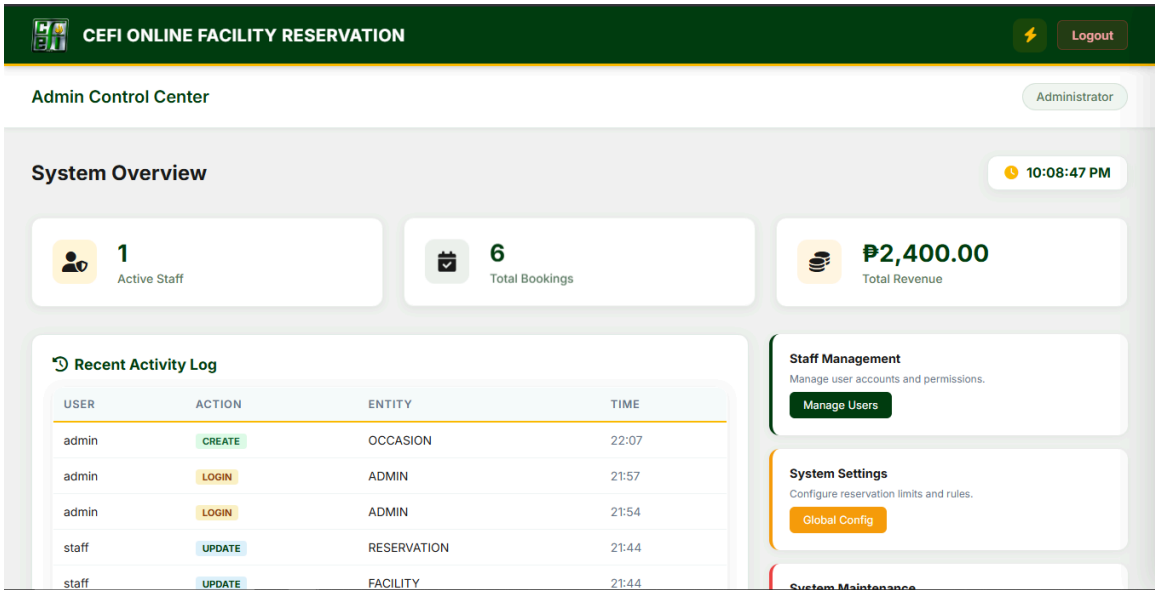
CORE MODULES

1. Login page



This is a single login page used by both staff and admin users. Everyone enters their username and password in the same form, and the system checks their account details. After logging in, it identifies whether the user is a staff member or an admin and then redirects them to the appropriate dashboard. In short, it is one shared login page with different access depending on the user’s role.

2. Admin dashboard



CEFI ONLINE FACILITY RESERVATION

Logout

System SettingsAdministrator

Reservation Rules

Max Reservations Per Day

Per user, per day limit

5

Max Advance Booking

Days ahead users can book

30

Min Advance Days

Minimum days before reservation date

2

Min Duration

Minimum hours per booking

1

Max Duration

Maximum hours per booking

8

Auto-Expire Pending

Hours before pending reservations expire

48

Allow Weekend Bookings

Enable Saturday/Sunday reservations

☒

System Configuration

System Name

Changed in headers and emails

CEFI Online Facility Reserva

Session Timeout

Minutes of inactivity before logout

30

Default Open Time

Facility default opening hour

07:00 AM

Default Close Time

Facility default closing hour

08:00 PM

Maintenance Mode

Disable public reservations

☐

Notifications

Email Notifications

Send email alerts for new reservations

☐

Admin Email

Receives notification emails

admin@cefi.edu

Reset

Save Settings

CEFI ONLINE FACILITY RESERVATION

Logout

System MaintenanceAdministrator

272.0 KB

Database Size

62

Audit Log Entries

6

Total Reservations

0

Backup Files

Backup Database

Create a full SQL backup of all tables and data. Backups are saved to the server's backups/ folder.

Create Backup

Clear Audit Logs

Remove old audit log entries to free up database space. Currently 62 entries since Apr 03, 2026.

Clear Logs

Optimize Database

Run table optimization to reclaim unused space and improve query performance across all tables.

Optimize Tables

Purge Old Reservations

Permanently delete all EXPIRED and CANCELLED reservations from the database.

Purge Records

Database Tables

8 tables - 272.0 KB total

This admin dashboard provides different features that allow the administrator to control and manage the system. It includes modules such as Facility Management (view-only access), Event Calendar, Manage Occasions, Admin Tools, Staff Management, System Settings, Maintenance, System Logs, Reservation History, and Audit Trail. Each section serves a specific purpose, like monitoring activities, managing users, configuring system rules, and maintaining the database. On the right side, quick actions are available such as managing user accounts, adjusting system configurations, and performing maintenance tasks like backups or clearing logs. Overall, this dashboard gives the admin full access to oversee system operations and ensure everything runs smoothly.

3. Staff dashboard

CEFI ONLINE FACILITY RESERVATION

1

Facebook Reservations!

daniel paraiso

Test Lab · Apr 27

Apr 03, 9:33 PM

View All Pending →

Logout

Staff

10:11:11 PM

FRI, APR 3, 2026

Dashboard

Welcome, Staff

1

Total Facilities

1

Available Facilities

6

Total Reservations

1

Today's Reservations

Reservation Status Overview

1

Pending

3

Approved

1

Rejected

1

Cancelled

Ongoing Events

This staff dashboard has simple features to help manage reservations. It shows the number of facilities and reservations, including those for today. It also displays the status of reservations such as pending, approved, rejected, and cancelled. There is a notification section for new requests and a list of ongoing events. Overall, it helps staff easily monitor and handle reservation activities.

- Super Admin: Full access to System Settings, Facility Management, Audit Logs, and User Account creation.
- Staff/Receptionist: Access to Manage Approvals, Walk-In Reservations, and Event Calendar view. Restricted from deleting Audit Logs or altering Facility Pricing.
- Client (Messenger User): View-only access to availability and the ability to submit reservation requests through the AI interface.

4. Walk-In

CEFI ONLINE FACILITY RESERVATION

1

Logout

Walk-in Registration

Create a reservation for a person physically present. Walk-in reservations are set as pending upon creation.

Full Name *

Enter full name

User Type *

Select Type

Email Address

user@email.com

Phone Number

09171234567

Facility

Select Facility

Reservation Date

04/03/2026

Number of Attendees

How many people?

Start Time

10:12 PM

End Time

This screen shows part of a **walk-in reservation form** in your system. It allows staff to enter details for a person who did not book in advance. The visible fields include **email address** and **phone number**, which are needed to record the guest’s information. This form is used when someone arrives personally, and the staff creates the reservation for them directly in the system.

5. Add Facility



CEFI ONLINE FACILITY RESERVATION





Logout

Add Facility

Facility Name

e.g. Gym, Multi-purpose Hall

Max Capacity

Number of people

Description

Describe the facility's amenities and rules...

Price per Hour (PHP)

0.00

Price per Day (PHP)

0.00

Opening Time

07:00 AM

Closing Time

08:00 PM

Advance Booking Days Required

2

Status

AVAILABLE

Min Duration (Hours)

Max Duration (Hours)

This page is used to **add a new facility** to the system. The admin or staff can enter details such as the **facility name**, **maximum capacity**, and a **description** of the place. It also includes pricing information like **price per hour** and **price per day**, as well as the **opening and closing time**. Additional settings like **advance booking days**, **status (available or not)**, and duration limits can also be set.

6. Manage Approvals



CEFI ONLINE FACILITY RESERVATION





Logout

All Reservations

+ Create Reservation

+ Walk-in Reservation

Export CSV

Search by name...

All Statuses

All Facilities

mm/dd/yyyy

mm/dd/yyyy

Filter

Clear

Showing 1-6 of 6 reservations

NAME	FACILITY	PURPOSE	DATE	TIME	COST	STATUS	ACTION
Carlo Reyes WALK-IN	Test Lab	sdaada	2026-04-03	07:00 - 15:00 8.0 hrs	₱1,800.00	APPROVED	Print Cancel Edit Delete
daniel paraiso	Test Lab		2026-04-27	07:00 - 15:00 8.0 hrs	₱800.00	PENDING 🕒 23h 17m left	Approve Reject Cancel Edit Delete
gian pena WALK-IN	Test Lab	asdsada	2026-04-30	15:00 - 17:00 2.0 hrs	₱200.00	CANCELLED Reason: gian pena request a cancellati...	Edit Delete

This page is used to **manage reservation approvals**. It allows staff to review reservation requests and decide whether to **approve, reject, or cancel** them. Each request shows details like the facility, time, duration, price, and reason. There are also action buttons (such as edit or delete) to update or manage the request.

7. Event Calendar

CEFI ONLINE FACILITY RESERVATION



1



Logout

Calendar

Admin

Reservation Calendar

All

Reservations

Occasions

< April 2026 >

Today

Approved

Pending

Rejected

Holiday

School Event

SUN	MON	TUE	WED	THU	FRI	SAT
29	30	31	1	2 ★ Maundy Thursday	3 ★ Good Friday 7:00 AM Test Lab 7:00 AM Test Lab	4 ★ Black Saturday
5	6 7:00 AM Test Lab	7	8	9 ★ Araw ng Kagitingan (Day of Valor)	10	11

This is the **event calendar feature** of the system. It allows users to view both **events and reservations** in a calendar format. Different colors are used to show the status, such as approved, pending, rejected, holidays, and school events. Users can easily check scheduled activities, available dates, and upcoming reservations.

8. Reservation History

CEFI ONLINE FACILITY RESERVATION



1



Logout

Reservation History

Status

All Statuses

Date

mm/dd/yyyy

Filter

Reset

DATE	FACILITY	USER / FB NAME	TIME	PURPOSE	STATUS	DATE CREATED	ACTIONS
Apr 30, 2026	Test Lab	gian pena WALK-IN	03:00 PM - 05:00 PM	asdsada		Apr 03, 2026 09:19 PM	Edit
Apr 30, 2026	Test Lab	Carlo Reyes WALK-IN	07:00 AM - 03:00 PM	asdasd	APPROVED	Apr 03, 2026 09:07 PM	Edit
Apr 27, 2026	Test Lab	daniel paraíso	07:00 AM - 03:00 PM		PENDING	Apr 03, 2026 09:33 PM	Edit
Apr 06, 2026	Test Lab	justine rovira WALK-IN	07:00 AM - 03:00 PM	asdasd	APPROVED	Apr 03, 2026 01:46 PM	Edit
Apr 03, 2026	Test Lab	Ana Villanueva WALK-IN	07:00 AM - 03:00 PM	asdsaj	REJECTED	Apr 03, 2026 09:04 PM	Edit
Apr 03, 2026	Test Lab	Carlo Reyes WALK-IN	07:00 AM - 03:00 PM	sdasda	APPROVED	Apr 03, 2026 09:43 PM	Edit

This is the **reservation history feature** of the system. It shows a list of all past and current reservations, including details like the date, facility, user, time, purpose, and status. Users can also filter the records by status or date to find specific reservations easily. Each entry shows whether it is approved, pending, or rejected, and includes an option to edit the record.

To prevent 'ghost bookings' and optimize facility usage, the system implements an Automated Expiry Policy. When an online reservation is made, the database sets a `verification_deadline` (typically 24-48 hours). If a student fails to present their requirements or confirm the booking within this window, the system automatically transitions the status from PENDING to EXPIRED, instantly releasing the time slot back to the calendar for other users.

9. Audit Trail

CEFI ONLINE FACILITY RESERVATION

Logout

Audit Trail

Staff

Action Type

All Actions

Entity Category

All Entities

Apply Filters

Reset

#	TIMESTAMP	STAFF MEMBER	ACTION	TARGET ENTITY	ACTIVITY DETAILS
1	Apr 03, 2026 22:10:58	staff	LOGIN	STAFF #1	staff logged in: staff
2	Apr 03, 2026 22:07:06	admin	CREATE	OCCASION #18	Added special occasion: Christmas
3	Apr 03, 2026 21:57:02	admin	LOGIN	ADMIN #2	admin logged in: admin
4	Apr 03, 2026 21:54:55	admin	LOGIN	ADMIN #2	admin logged in: admin
5	Apr 03, 2026 21:44:23	staff	UPDATE	RESERVATION #6	Updated reservation for Carlo Reyes (Status: APPROVED) — Reason: dasdsadsada
6	Apr 03, 2026 21:44:13	staff	UPDATE	FACILITY #4	Updated facility: Test Lab

This is the **audit trail feature** of the system. It records all actions done by users, such as logins, creating records, and updating reservations or facilities. Each entry shows the time, user, action, and details of what was done. There are also filters to search specific activities.

10. Facebook Messenger / n8n Integration Flow

Meta

My Apps

Required actions

Docs

Tools

Support

Search...

n8nmessenger

App ID: 1608667137119051

App Mode: Development

Live

App type: Business

Help

Dashboard

Required actions

App settings

App roles

Alerts

App Review

Products

Webhooks

Messenger

Messenger API Settings

API integration helper

Instagram settings

Activity log

Activity log

you will be able to send messages to anyone that has been granted either the administrator, developer or tester role for your app.

To get started, please read the [developer documentation](#) for complete details. Your access to and use of the platform are subject to [Platform Policies](#) and [Developer Policies](#), together with all other applicable terms and policies.

Increase traffic to Messenger

Create ads to help more people discover your experience in Messenger.

Learn more

1. Configure webhooks

Configure a custom webhook URL or use services that help you set up an endpoint. [Learn more.](#)

Callback URL

https://reusable-paulita-reductional.ngrok-free.dev/webhook/689d22e7-434b-4d08-95d7-758443c11c68

Verify token

Attach a client certificate to Webhook requests. [Learn more.](#)

Webhook fields

Field	Version	Test	Subscribe
affiliation	v25.0	Test	Unsubscribed
attire	v25.0	Test	Unsubscribed
awards	v25.0	Test	Unsubscribed

Inbox

Respond to messages, set up automations and more.

Create messaging ad

Available

All messages

Messenger

Instagram

WhatsApp

Facebook comments

Instagram comments

Search

Manage

Unread

Priority

Ad replies

Follow up

Markwilson De Torres

You: Language set successfully! Paano ko po...

Wed

Justine Lusterio Rovira

Hello

Tue

Mafe Merle

Magpa book ulit ako

Tue

Paul Lavarez

Hello

Tue

Monica Amparo-Ramchand

Ano ang mga available facility?

Tue

Markwilson De Torres

Assign this conversation

English

Pure Tagalog

Taglish

English

Language set successfully! Paano ko po kayo matutulungan / How can I help you book today?

Reply in Messenger...

Contact details

Add more details about this contact.

Add details

Facebook profile

Public info sourced from Markwilson's profile

Local time 12:47 PM

Studied at LCNHS

Lives in Lucena City

From Lucena City

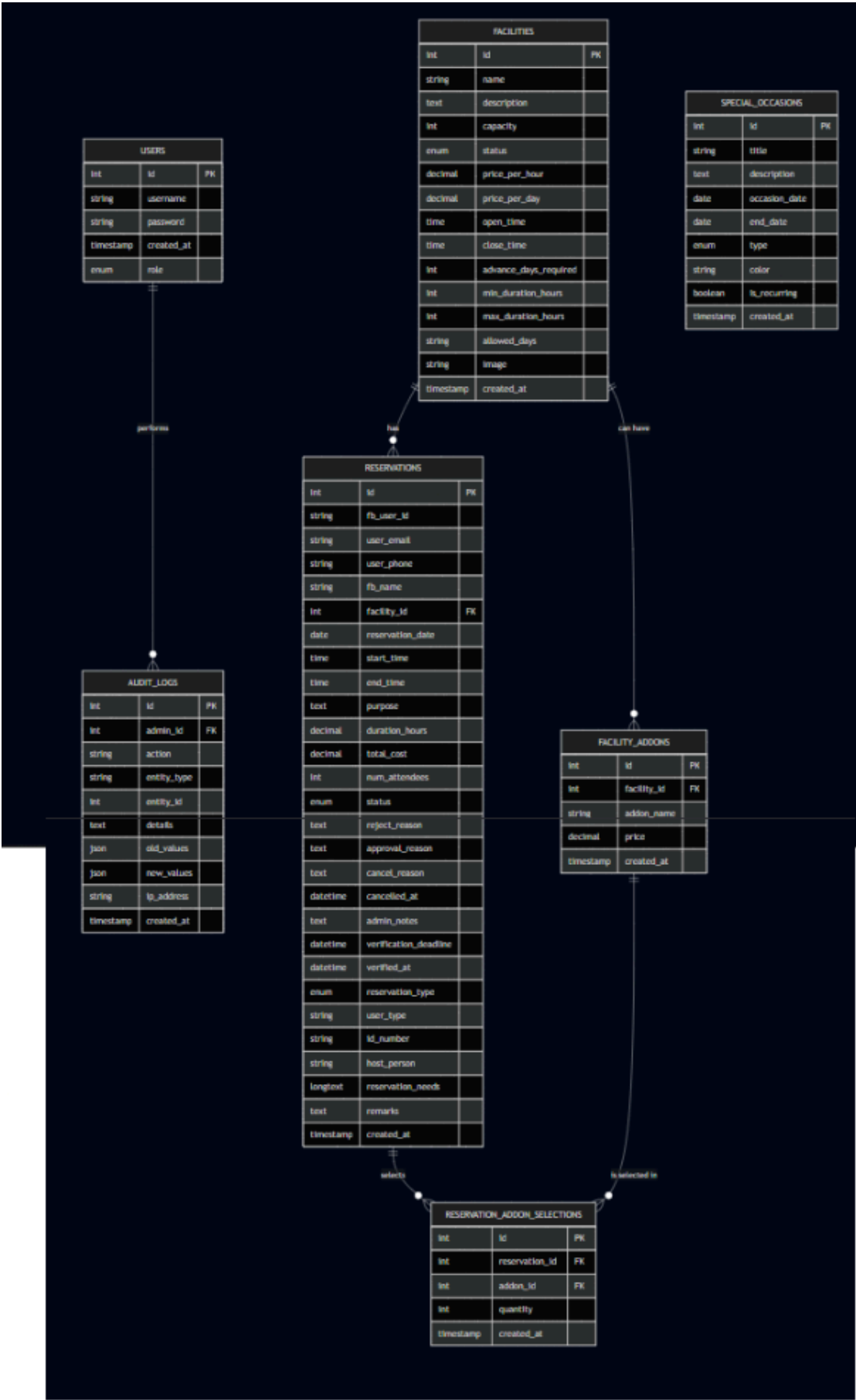
Activity

Recommended

Marking activities like orders and leads can improve ad performance.

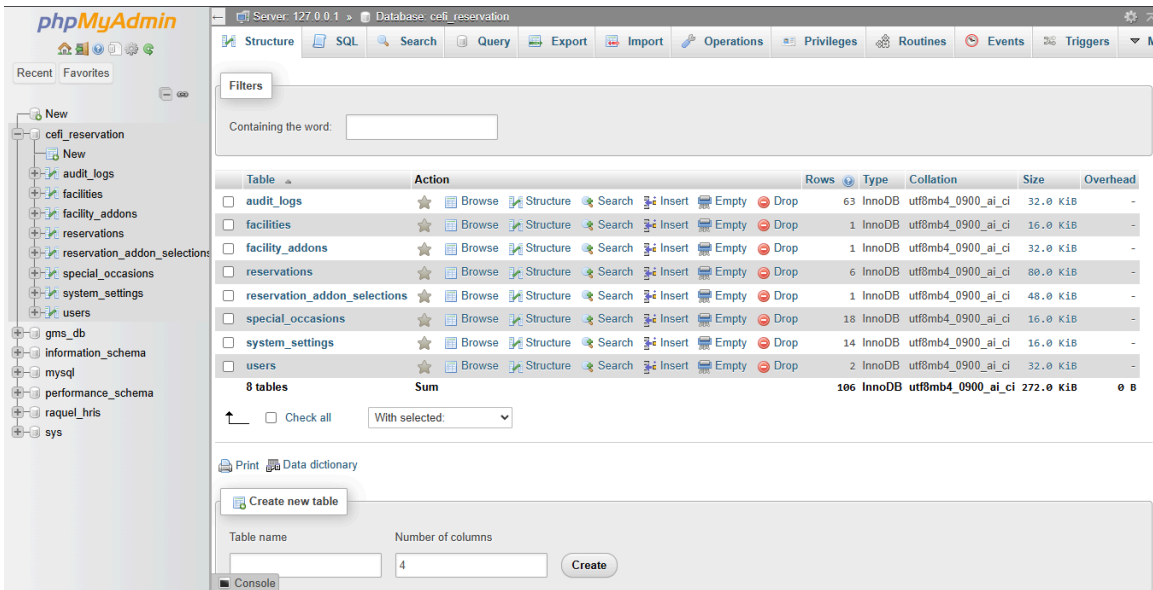
Order status

Create order



reservation_addons_selection table, allowing multiple add-ons per booking. The system also includes a **special_occasions** table to manage holidays or blocked dates that may affect availability, and an **audit_logs** table to track system activities and changes for monitoring purposes. Overall, the diagram shows how users can book facilities, optionally add services, and go through an approval and tracking process within a structured database system.

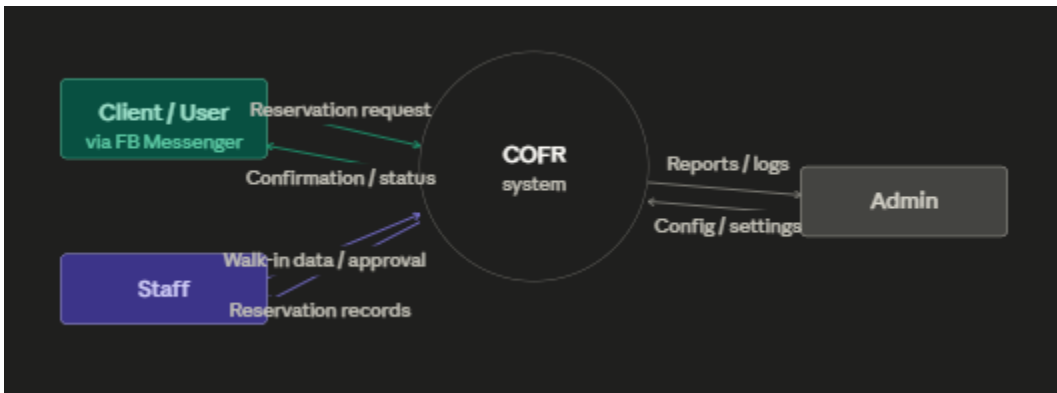
- Screenshots of database management tools

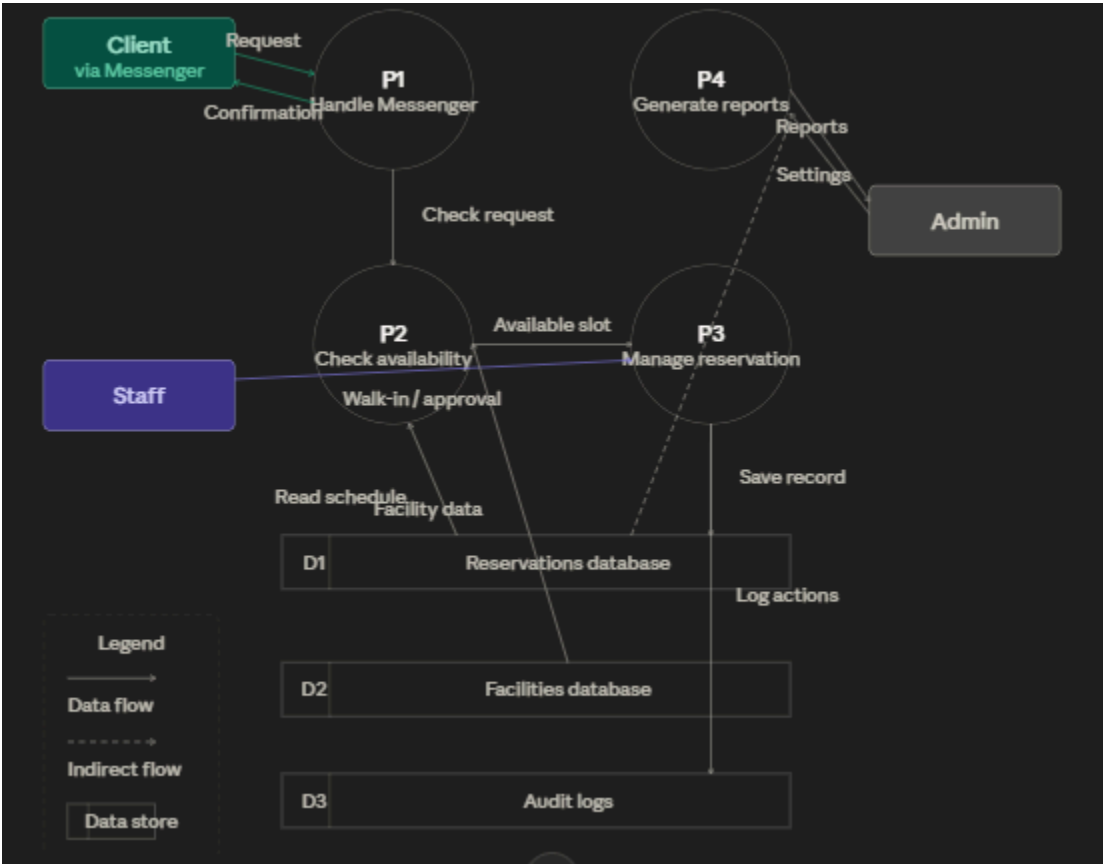


The screenshot displays the phpMyAdmin interface for the 'cefl_reservation' database. The left sidebar shows the database structure with tables like audit_logs, facilities, reservations, and reservation_addons_selection. The main panel shows the 'Structure' tab for the 'users' table, listing columns and their types. Below the table list, there is a 'Create new table' form with fields for 'Table name' and 'Number of columns'.

Table	Action	Rows	Type	Collation	Size	Overhead
audit_logs	Browse Structure Search Insert Empty Drop	63	InnoDB	utf8mb4_0900_ai_ci	32.0 K1B	-
facilities	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	16.0 K1B	-
facility_addons	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	32.0 K1B	-
reservations	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_0900_ai_ci	80.0 K1B	-
reservation_addon_selections	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_0900_ai_ci	48.0 K1B	-
special_occasions	Browse Structure Search Insert Empty Drop	18	InnoDB	utf8mb4_0900_ai_ci	16.0 K1B	-
system_settings	Browse Structure Search Insert Empty Drop	14	InnoDB	utf8mb4_0900_ai_ci	16.0 K1B	-
users	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_0900_ai_ci	32.0 K1B	-
8 tables	Sum	106	InnoDB	utf8mb4_0900_ai_ci	272.0 K1B	0 B

d. Project Diagram (Ex: Data Flow Diagram, Use Case Diagram)





The system follows a Hybrid Cloud Architecture. The Presentation Layer consists of the Facebook Messenger UI and a PHP-based Admin Dashboard. The Logic Layer is driven by n8n (automation) and AI models for natural language understanding. The Data Layer uses a MySQL database hosted on a local CEFI server, ensuring that all sensitive student data and audit trails remain within the institution's network infrastructure while utilizing cloud-based AI for user interactions.

- IV. Student Learning & Reflection
- a. Individual Reflection

b. Skills Gained

c. Challenges Encountered

d. Lessons Learned
- V. Project Evaluation
- Client Evaluation Form

- Adviser/Consultant Evaluation Rubric

- Panel Evaluation Results

CLIENT FEEDBACK FORM

Project Title:	CEFI Facebook Messenger-Based Facility Reservation System with AI Automation using n8n
Members:	De Torres, Markwilson Rovira, Justine
Consultant:	Paul Lavarez
Business Name:	Calayan Educational Foundation Inc.
Address:	Maharlika Highway Ibabang Dupay, Red-V Lucena City, Philippines
Client's Name:	



Instructions: Please provide honest feedback on the project, its deliverables, and the team’s performance. Your feedback is valuable to improve the quality of the IT solution.

Criteria: (4) Excellent | (3) Good | (2) Satisfactory | (1) Needs Improvement

	4	3	2	1
System Functionality				
System meets the business requirements				
Features are user-friendly				
Reliability and performance				
System Design and Interface				
Interface design is clear and intuitive				
Navigation is easy and logical				
Documentation and Training				
User manual or guide provided				
Adequate training/support given				
Overall Satisfaction				
Overall satisfaction with the project outcome				
Likelihood of recommending or continuing use				

Additional Comments / Suggestions

Client
Signature: _____

Date
: _____ 4/7/2026